in fibre optic systems and other industrial and consumer applications.

Antenna gain is inversely proportional to the wavelength of operation. So, operating in lower wavelength bands is beneficial. But eye-safety regulations allow transmitting much more power in higher wavelengths.

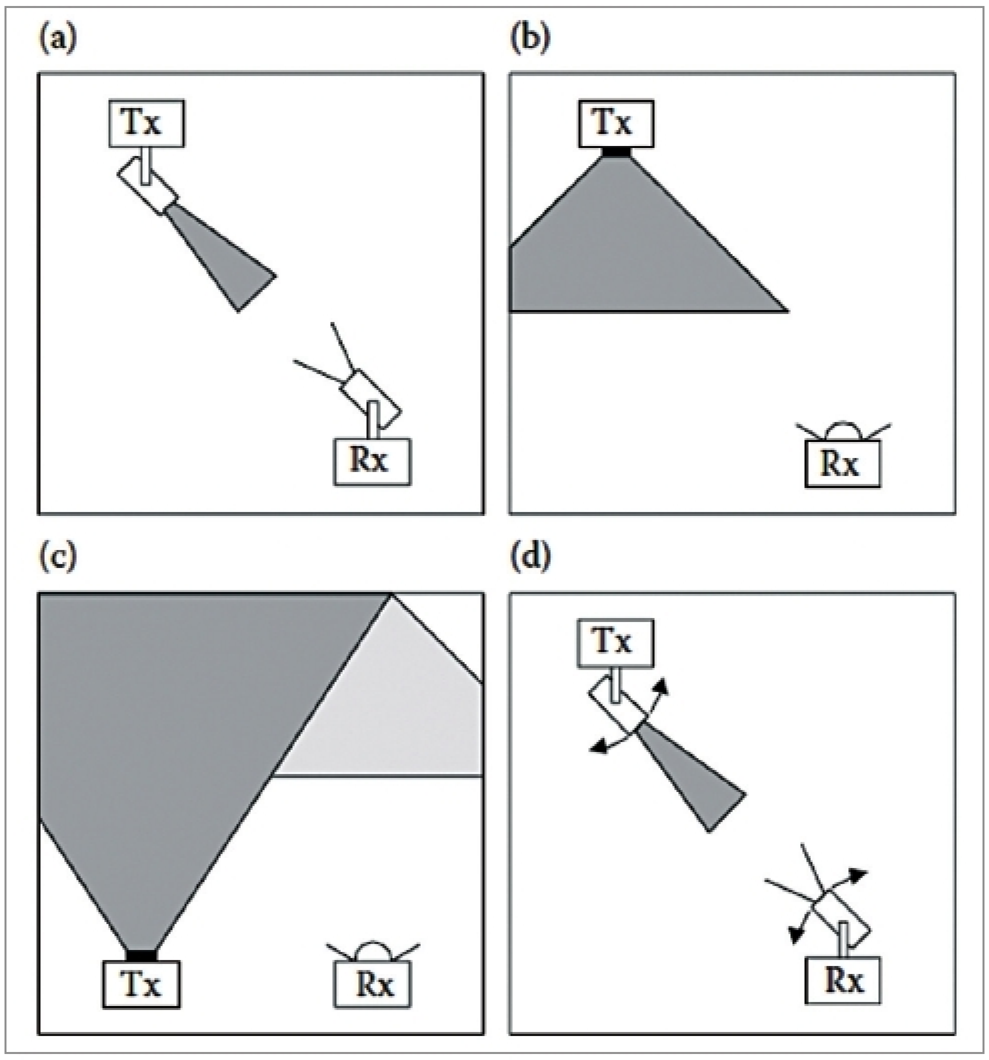

Fig. 2: Indoor WOC System: (a) Directed LOS, (b) Non-directed LOS, (c) Diffused, (d) Tracked (Source: Optical Wireless Communications, CRC Press)

Advantages of WOC system

The WOC system has several advantages over RF communication system due to the shorter wavelength of operation.

Higher bandwidth. An increase in carrier frequency increases the information-carrying capacity of a communication system. In optical communication, the usable bandwidth is almost 10^5 times of that in the RF communication system.

Unlicensed spectrum. Till now, there is no licensing requirement for the establishment of an FSO communication link. It makes the development and deployment of an FSO communication system easier.

Better security. The beam divergence is proportional to the wavelength. As in the case of WOC communication, the operating wavelength is much less than of RF communication, and the beam divergence is significantly less. As the beam is confined in a narrow region, it provides better security and power efficiency.

Less mass and power. Due to narrow beam divergence, the received power intensity is higher for a given transmitted power. It allows a simpler and lighter transmitter and receiver design. Antenna gain is inversely proportional to the square of the operating wavelength. A small antenna in the WOC system can achieve higher gain.

Disadvantages of WOC system

LOS requirement. Due to a very narrow beam divergence, LOS is a critical requirement for successful communication.

Pointing and tracking requirement. Again, due to narrow beam divergence, even building vibrations pose a severe issue to the establishment of a good communication link. Accurate pointing and tracking systems need to be deployed for reliable operation.

Atmospheric effects. WOC communication is highly impacted by haze and fog due to the high attenuation of optical signals. Random temperature variations in day time cause random variation of the refractive index of the air. This atmospheric turbulence causes random intensity variation in the receiver.

Background noise power. The relative position of the sun with respect to the transmitter and receiver plays a crucial role. Sun is a potential source of background noise. Under certain conditions, background noise may severely impair a WOC link.

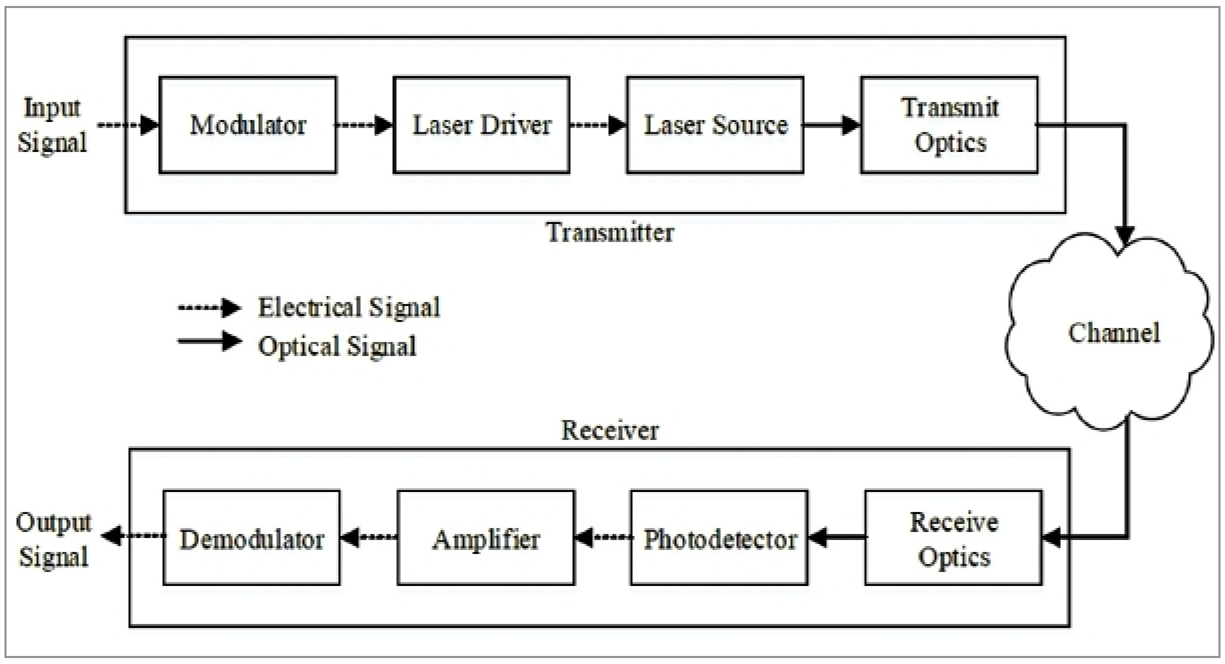

Fig. 3: Block diagram of a WOC system

Applications

WOC communication has versatile applications. Key application areas are:

- Building to building communication
- Disaster recovery applications
- Satellite and deep space links
- As a backup to an existing optical fibre link
- X-haul to 3G, 4G, and 5G mobile communication system

The FSO system is gaining rapid popularity for Gbps capable communication link. The research works on WOC system focus to mitigate the atmospheric effects and enhance throughput under various atmospheric conditions. The use of sophisticated forward error correction techniques and aperture averaging are some of the enablers for a robust communication link. EFY